

LAB 1: Environment Setup & Core PHP Basics

- **Syllabus Alignment:** Unit 1: Introduction (10 LHs) — Server-side Scripting, Development Server, Syntax, Comments, Output
- **Lab Objectives:** To install a local server stack (Apache, PHP, MySQL), manage configurations, and understand the basic request-response cycle using fundamental parsing markers.
- **Lab Questions:**
 1. **Environment Setup:** Install a localized development server engine (such as XAMPP). Create a file named `info.php` containing the `phpinfo()`; debugging function, execute it via `localhost`, and verify that the Apache server and PHP extensions are properly configured.
 2. **Basic Output & Comments:** Write a PHP script that renders a basic welcoming message to the browser using the `echo` statement. Incorporate single-line (`//` and `#`) and multi-line (`/* ... */`) comments within the source code to document your setup steps.

Lab Report Skeleton: Lab 1

1. Objectives

- [Task] Document the setup of the local development architecture (XAMPP).
- [Task] Outline the verification process for the local server and the PHP engine.
- [Task] List the core rules of PHP delimiters, basic string outputs, and script comments.

2. Theoretical Introduction

- **Server-Side Scripting:** Define how server code compiles on the backend before sending HTML down to the browser node.
- **Development Stack Components:** Briefly define the roles of Apache (Web Server), PHP (Interpreter), and MySQL (Database).
- **Lexical Rules:** Outline what `<?php ... ?>` tags do, how `echo` outputs content, and how comments (`//`, `#`, `/* */`) are treated by the compiler.

3. Requirements & Tools Used

- **Hardware Platform:** [Insert CPU, RAM, and OS used, e.g., Windows 11]

- **Software Stack:** XAMPP Server Package, Text Editor/IDE (e.g., VS Code), and a Web Browser.

4. Step-by-Step Procedure

1. [Step] Launch and toggle the **Apache Web Server** module active inside the local environment panel.
2. [Step] Navigate to the root web directory deployment workspace (`htdocs` folder).
3. [Step] Create new plaintext documents, add your scripting logic, and save files explicitly with the `.php` extension.
4. [Step] Open the web browser application and test your scripts using the localized network protocol pathway: `http://localhost/{filename}.php`.

5. Lab Assignments & Source Code

◆ Assignment 1: Environment Verification

- **Problem Statement:** Write a script using structural debugging lookups to verify runtime configurations.
- **Source Code Box:**

6. Results & Output Previews

- **Output 1 (`info.php`):** [Describe or paste a clear screenshot of the rendered system configuration tables]
- **Output 2 (`hello.php`):** [Provide the exact plain text string output rendered by your script in the browser window]

7. Conclusion

- [Final Note] Write a summary confirming the local environment was successfully initialized.
- [Final Note] State your final takeaway regarding how server-side processing operates and how basic language syntax structures behave.